

RECOMMENDATIONS FOR ADAPTATION, MITIGATION AND INTEGRATING CLIMATE-INDUCED HAZARD & VULNERABILITY CONCERNS INTO SECTORAL DEVELOPMENT STRATEGY AND PLANS

July 2026

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1. About the Report

Climate change is increasingly altering the frequency, intensity and spatial distribution of climate-induced hazards across Uttarakhand, with significant implications for ecosystems, infrastructure, livelihoods and socio-economic development. Rising temperatures, changing precipitation patterns, glacier retreat, increasing occurrence of extreme rainfall, droughts, forest fires, heat stress, landslides, flash floods and Glacial Lake Outburst Floods (GLOFs) are creating complex and interconnected risks that affect multiple sectors simultaneously. These changes are expected to intensify under future climate scenarios, increasing exposure, amplifying existing vulnerabilities and placing additional pressure on natural resources, public infrastructure and essential services.

Recognising the need for evidence-based adaptation planning, the Government of Uttarakhand, through the State Environment Conservation and Climate Change Directorate (SECCCD) with support from GIZ, is undertaking a comprehensive Climate Vulnerability Assessment covering all 95 development blocks across 13 districts. The assessment generates multi-hazard assessment, sectoral climate impact analysis, vulnerability assessment and climate risk assessment to generate spatially explicit evidence for climate-resilient planning and decision-making. The assessment covers seven priority sectors: Agriculture, Forest and Biodiversity, Water Resources and WASH, Health, Energy, Roads and Linear Infrastructure, and Tourism and will culminate in the development of a Digital Climate Vulnerability Atlas for Uttarakhand.

This report, **Recommendations for Adaptation, Mitigation and Integrating Climate-Induced Hazard Vulnerability Concerns into Sectoral Development Strategies and Plans**, constitutes the adaptation planning component of the broader vulnerability assessment. It translates the findings of climate impact, vulnerability and risk assessments into a structured portfolio of adaptation measures and policy recommendations that can inform sectoral planning and public investment. The report is intended to support the integration of climate risk considerations into development planning at the State, district and block levels, thereby strengthening climate resilience across sectors.

The recommendations presented in this report are informed by the analytical framework adopted under the vulnerability assessment. Sector-specific impact pathways are subsequently developed to identify the direct and indirect consequences of these hazards on socio-economic systems, infrastructure and ecosystem services. Vulnerability is assessed through the interaction of hazard exposure, sensitivity and adaptive capacity, while climate risk is evaluated through the integration of hazard, exposure and vulnerability in accordance with the IPCC risk framework. The resulting evidence provides the basis for identifying adaptation priorities and resilience-building interventions across sectors.

Rather than addressing individual hazards in isolation, the recommendations recognise that climate risks often arise through cascading interactions across sectors and scales. **From hazard mapping to adaptation needs**, through block-level vulnerability scoring to a Composite Risk Index and then derives specific adaptation needs across five vulnerability dimensions: Environmental, Social, Institutional, Economic and Physical, using a structured impact-to-need pathway presented as a single reference diagram.

Three findings recur across sectors and justify the structure of the recommendations that follow. First, a small set of blocks: Joshimath, Dasholi, Bhatwari, Ukhimath, Kapkot, Dharchula and Munsyari foremost among them, carry compounding, multi-sector exposure: the same terrain that produces landslide and flood risk for roads and water schemes also destabilises hydropower assets, disrupts health-facility access, and undercuts orchard and transhumance livelihoods. Adaptation measures are therefore organised around the principal drivers of vulnerability and are designed to reduce exposure, minimise sensitivity and enhance adaptive capacity. The proposed interventions encompass structural and non-structural measures, ecosystem-based adaptation, climate-resilient infrastructure, institutional strengthening, improved climate information services, technological innovation, nature-based solutions, governance reforms and community-based adaptation approaches.

Sector-specific recommendations are presented for Agriculture, Forest and Biodiversity, Tourism, Water Resources and WASH, Roads and Linear Infrastructure, Energy, and Health. For each sector, the report identifies the principal

climate risks, adaptation objectives, strategic interventions and enabling actions required to enhance resilience under current and projected climate conditions. Recognising the substantial spatial heterogeneity of climate risks across Uttarakhand, the report further provides **block-level adaptation priorities for all 95 development blocks**, enabling geographically differentiated planning and targeted allocation of resources.

Given that adaptation is fundamentally a development process, the report places particular emphasis on the mainstreaming of climate risk considerations into existing planning and governance mechanisms. It provides guidance for integrating climate resilience into State, district and block development plans, sectoral strategies, infrastructure investment decisions, land-use planning, watershed management, disaster risk reduction programmes and local development planning processes. The report also identifies opportunities for convergence with ongoing national and State programmes to facilitate implementation through existing institutional and financial mechanisms rather than creating parallel structures.

The report further proposes an implementation roadmap supported by measurable monitoring indicators to facilitate the tracking of adaptation progress over time. The indicators are aligned with the vulnerability assessment framework and are intended to support periodic updates of vulnerability profiles, assessment of adaptation effectiveness and continuous refinement of policy interventions. This adaptive management approach is expected to strengthen the State's capacity to respond to evolving climate risks under changing socio-economic and climatic conditions.

Overall, this report provides a strategic framework for translating climate risk information into actionable adaptation planning. By linking scientific evidence with policy and planning processes, it seeks to support the systematic integration of climate resilience into sectoral development pathways and strengthen the long-term adaptive capacity of Uttarakhand's ecosystems, infrastructure, institutions and communities in the face of increasing climate variability and change.

2. Progress of Developing Vulnerability Assessment in the Process to Formulate Specific Adaptation Measures

2.1 Overview

The Climate Vulnerability Assessment undertaken for Uttarakhand is intended not only to identify areas and sectors at greatest risk from climate change, but also to provide an evidence base for adaptation planning. Consistent with the IPCC AR6 framework, adaptation planning is an iterative process in which climate information, impact assessment, vulnerability assessment and risk analysis are translated into adaptation responses that reduce climate risk while supporting sustainable development.

This report represents the adaptation planning component of the overall assessment. It builds upon the completed climate hazard, impact, vulnerability and risk assessments to identify sector-specific and block-specific adaptation measures that address the underlying drivers of vulnerability and strengthen long-term resilience.

The chapter describes the analytical progression from climate projections to adaptation planning, providing the basis for the sectoral recommendations presented in Chapter 3 and the block-wise adaptation measures presented in Chapter 4.

2.2 Assessing Impact, Vulnerability and Risk

The formulation of adaptation measures builds upon the climate risk assessment completed under the Uttarakhand Climate Vulnerability Assessment. The assessment integrates downscaled climate projections developed by IIT Roorkee with multi-hazard mapping, sectoral impact assessment and block-level vulnerability assessment to understand how future climate change may affect people, ecosystems, infrastructure and economic activities across Uttarakhand.

The assessment was undertaken through a sequential analytical framework comprising:

Step 1. Climate Hazard Assessment

Historical observations and future climate projections (SSP2-4.5 and SSP5-8.5) were analysed to characterise changes in temperature, precipitation and climate extremes. Multi-hazard assessments were subsequently undertaken for landslides, floods, droughts, forest fires, heat stress, extreme rainfall, Potentially Dangerous Glacial Lakes (PDGLs) and other climate-induced hazards.



Step 2. Climate Impact Assessment

Sector-specific climate impact pathways were developed to understand how projected climate hazards affect environmental systems, infrastructure, livelihoods, public services and socio-economic development across Agriculture, Forests, Water Resources, WASH, Roads, Energy, Tourism and Health.



Step 3. Vulnerability Assessment

Sector-specific vulnerability indices were developed using indicators representing **hazard exposure, exposure, sensitivity and adaptive capacity**. Block-level assessments were undertaken for Agriculture (AVI), Water

Resources (WVI), Roads (RVI), Health (HVI) and Energy (EVI), and subsequently integrated into a Composite Vulnerability Index (CVI).



Step 4. Climate Risk Assessment

Sectoral vulnerability was combined with hazard probability following the IPCC AR6 risk framework to generate block-level climate risk profiles for all 95 development blocks. These assessments identify where climate risks are expected to be greatest under both current and future climate conditions.

The table below summarises the state-level climate projections underpinning every sector assessment in this report.

Parameter	Baseline (1981–2014)	Mid-century (2021–2050)	End-century (2071–2100)
Temperature rise		+1.3 to 1.6°C	+2.3 to 5.9°C
Monsoon precipitation changes		Modest increase	+6 to 16%, higher inter-annual variability
Pre-monsoon rainfall		Gradual decline	Continued decline
Glacier ice loss (2000–2014 observed)	~16 Gt (3.5% of volume, ~1,950 glaciers)	Accelerating melt	10–20% of annual runoff from glacial melt
Landslide risk increase (monsoon, select districts)	Baseline	Rising	Up to +90% (May–October)

2.3 Translating Climate Risk into Adaptation Needs

The objective of adaptation planning is not simply to respond to hazards, but to reduce unacceptable climate risk by addressing the underlying drivers of vulnerability. Accordingly, adaptation needs have been identified by analysing how projected climate impacts influence environmental systems, infrastructure, institutions, economies and communities, and by determining where interventions can most effectively reduce exposure, reduce sensitivity and strengthen adaptive capacity.

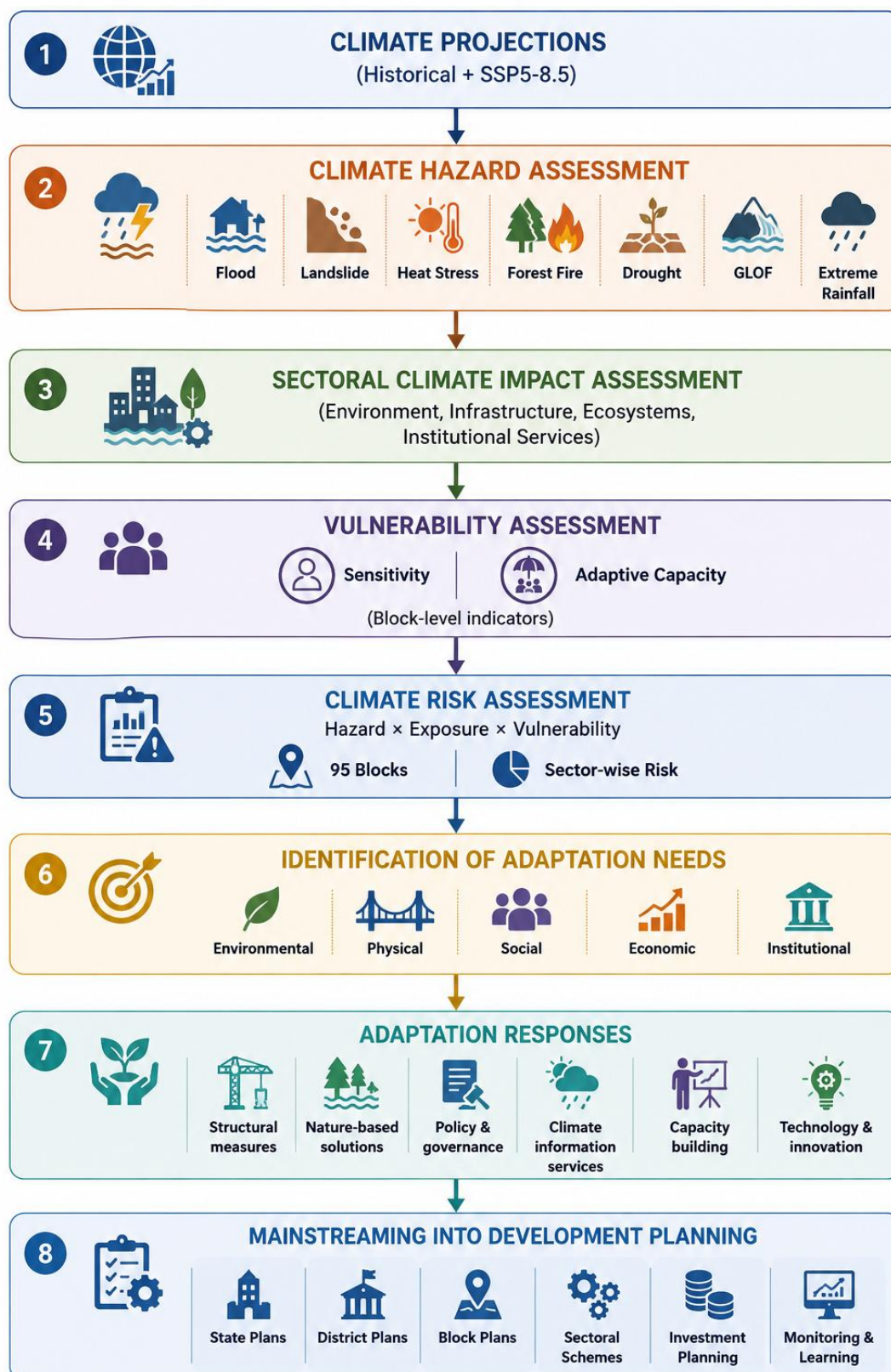
Rather than prescribing hazard-specific responses, this report adopts a systems-based approach in which adaptation needs are organised across five interrelated dimensions of vulnerability:

- Environmental
- Physical
- Social
- Economic
- Institutional

Each dimension represents a different pathway through which climate change influences development outcomes and therefore requires a distinct set of adaptation interventions. For example, ecosystem degradation may require ecosystem restoration, declining infrastructure reliability may require resilient engineering solutions, while institutional constraints may require strengthened governance, coordination and climate information systems.

These adaptation needs provide the organising framework for the sectoral recommendations presented in Chapter 3 and the block-specific adaptation matrix presented in Chapter 4.

Framework for Developing Adaptation Measures

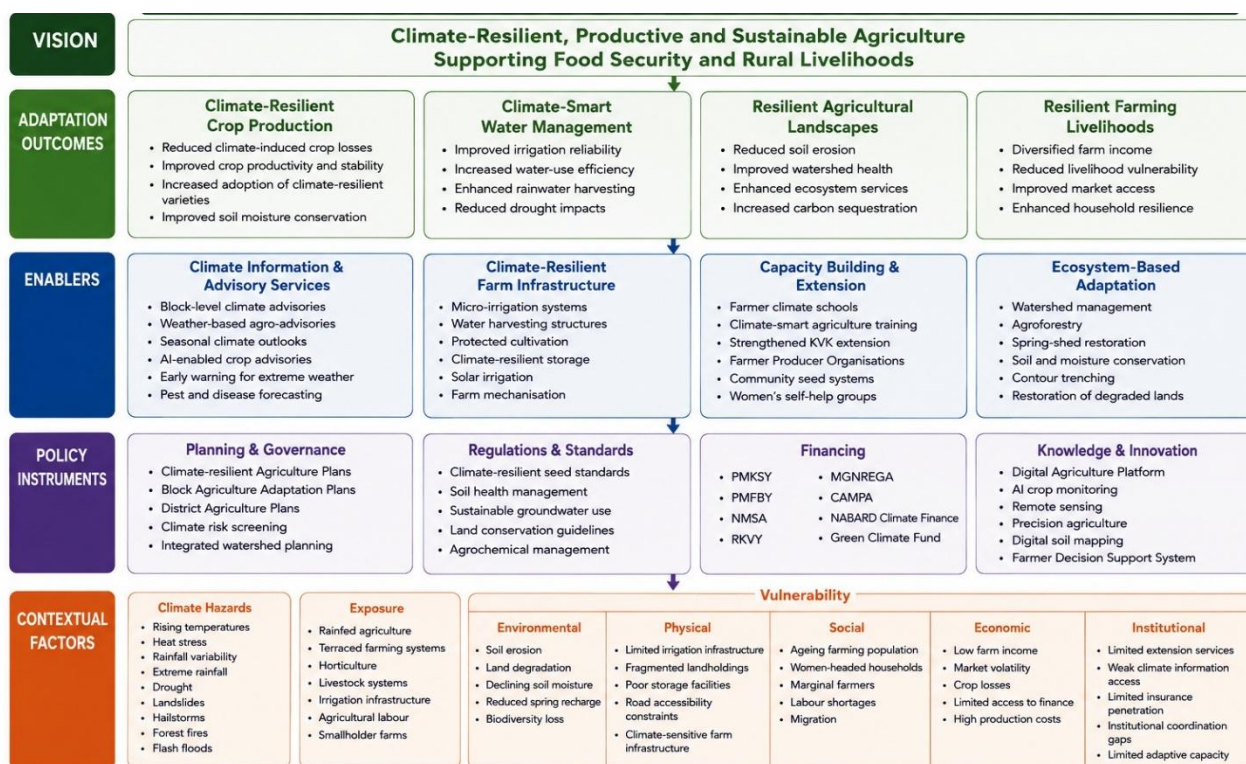


3. Sector-wise Recommendations

This section sets out adaptation and mitigation recommendations for each of the seven priority sectors assessed. Each sub-section summarises the sector's dominant climate exposure (drawn from the corresponding Sector Impact Assessment Report), then distinguishes established, already-proven measures suitable for immediate scale-up within existing institutional capacity from emerging and frontier measures that extend current practice using new technology, financing instruments or governance models. Block-specific application of these measures is operationalised in the Section 4 adaptation matrix.

3.1 Agriculture, Horticulture & Allied Livelihoods

Lead department(s): Department of Agriculture, Department of Horticulture, Department of Animal Husbandry & Dairy Development, Department of Fisheries, Uttarakhand State Rural Livelihood Mission (USRLM)



Adaptation Objective

Enhance the resilience of Uttarakhand's agricultural systems by strengthening climate-resilient production, improving water and soil resource management, safeguarding farming livelihoods, and promoting diversified and adaptive mountain agriculture under changing climatic conditions.

Adaptation Rationale

Agriculture in Uttarakhand is increasingly affected by both gradual climatic changes and extreme weather events. Rising temperatures and declining winter chilling hours are reducing the suitability of traditional temperate horticulture in several mid- and high-altitude districts, while increasing rainfall variability, prolonged dry spells and extreme precipitation events are affecting rainfed farming systems across the middle Himalayas. Cloudbursts, flash floods, hailstorms and landslides frequently damage terraced agriculture, irrigation channels and orchard infrastructure, resulting in recurring production losses. Simultaneously, changing land-use patterns and forest dynamics are contributing to increasing human-wildlife conflict, further reducing agricultural productivity and increasing livelihood risks in several mountain districts.

The Climate Impact Assessment indicates that districts dependent on horticulture, rainfed agriculture and fragile mountain landscapes are likely to experience increasing climate stress under future climate scenarios. Consequently, adaptation requires an integrated approach that combines climate-resilient production systems, improved water management, resilient infrastructure, ecosystem restoration, risk financing and climate-informed agricultural services.

Strategic Adaptation Measures

1. Climate-Resilient Crop and Horticulture Systems

- Promote climate-resilient crop varieties, including low-chilling horticultural cultivars, drought-tolerant cereals and disease-resistant planting material.
- Support diversification towards climate-resilient high-value crops including off-season vegetables, medicinal and aromatic plants, spices, mushroom cultivation and beekeeping.
- Encourage high-density orchards, protected cultivation, anti-hail protection systems and climate-resilient nursery development.
- Strengthen community seed systems and establish climate-resilient seed and fodder banks at block level.

2. Climate-Smart Water and Soil Management

- Expand micro-irrigation systems including drip and sprinkler irrigation in water-stressed agricultural areas.
- Restore traditional water harvesting systems, village tanks and khals, integrated with springshed management.
- Promote watershed development, soil and moisture conservation, contour bunding and terrace stabilization.
- Improve climate-resilient irrigation planning using integrated watershed approaches.

3. Climate-Resilient Farm Infrastructure

- Strengthen climate-resilient storage facilities, pack houses and cold-chain infrastructure.
- Protect orchard systems through anti-hail nets, windbreaks and resilient plantation design.
- Restore damaged terraces and irrigation channels following extreme weather events.
- Promote resilient farm mechanisation suitable for mountain agriculture.

4. Climate Risk Management and Farmer Support

- Strengthen weather-based crop advisory services and seasonal climate forecasting.
- Expand climate-responsive crop insurance and simplify claim settlement mechanisms.
- Implement integrated human–wildlife conflict management through solar fencing, early warning systems and community-based surveillance.
- Develop contingency crop planning for droughts, floods and extreme weather events.

5. Ecosystem-Based Adaptation for Agriculture

- Promote agroforestry and integrated farming systems.
- Restore degraded watersheds supporting agricultural landscapes.
- Enhance pollinator habitats and biodiversity-friendly farming practices.
- Improve soil organic carbon through conservation agriculture and sustainable land management.

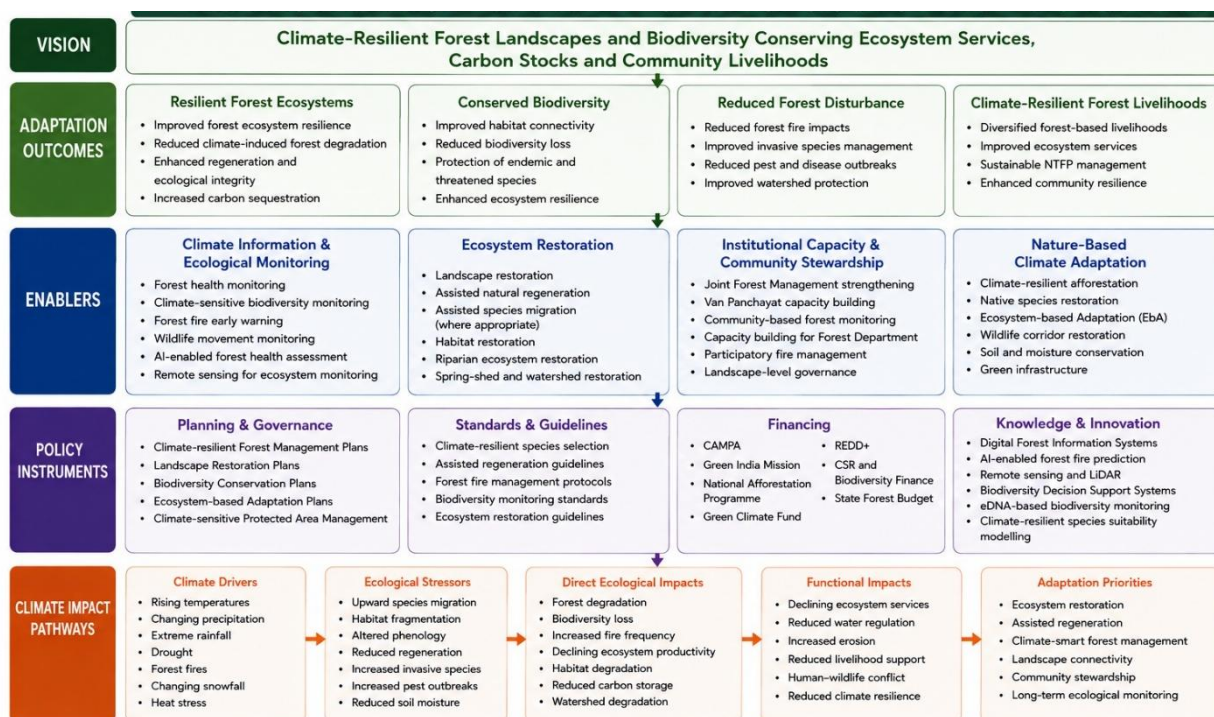
6. Climate Information, Innovation and Digital Agriculture

- Develop AI-enabled hyper-local agro-advisory systems for frost, hailstorms, pest outbreaks and extreme rainfall.
- Deploy remote sensing for crop health monitoring, orchard condition assessment and chilling-hour analysis.
- Establish digital agriculture platforms integrating weather forecasts, pest surveillance, irrigation advisories and market information.
- Apply geospatial decision-support systems to prioritise climate adaptation investments at block level.

7. Institutional Strengthening and Capacity Building

- Integrate climate risk assessment into district and block agricultural planning.
- Strengthen extension services for climate-smart agriculture through Krishi Vigyan Kendras and line departments.
- Build the capacity of Farmer Producer Organisations, Self-Help Groups and local institutions for climate adaptation.
- Promote convergence among the Departments of Agriculture, Horticulture, Animal Husbandry, Forests, Rural Development and Disaster Management for integrated adaptation planning.

3.2 Forest & Biodiversity



Adaptation Objective

Enhance the resilience of Uttarakhand's forest ecosystems and biodiversity to climate change by conserving ecosystem integrity, strengthening ecological connectivity, restoring degraded landscapes, reducing climate-induced disturbances, and sustaining ecosystem services and forest-dependent livelihoods.

Adaptation Rationale

Uttarakhand's forests provide critical ecosystem services, including water regulation, carbon sequestration, biodiversity conservation, slope stabilization and livelihood support for mountain communities. Climate change is altering the structure and functioning of these ecosystems through rising temperatures, changing precipitation patterns, declining snowfall, increasing frequency of forest fires, prolonged dry periods and extreme rainfall events. These climatic changes are contributing to shifts in species distribution, altered phenology, increased pest and disease incidence, habitat degradation, reduced natural regeneration and increasing fragmentation of ecologically important landscapes.

The Climate Impact Assessment indicates that climate change is likely to intensify ecosystem degradation and reduce the capacity of forests to provide essential ecosystem services, with cascading implications for water security, disaster risk reduction, biodiversity conservation and rural livelihoods. Adaptation therefore requires integrated landscape management that combines ecosystem restoration, biodiversity conservation, community participation and climate-informed forest governance.

Strategic Adaptation Measures:

1. Climate-Resilient Forest Landscape Restoration

- Restore degraded forest landscapes through native species plantation and assisted natural regeneration.
- Prioritise restoration of degraded catchments, riparian forests and upper watershed ecosystems.
- Promote climate-resilient species mixtures rather than monoculture plantations.
- Strengthen ecological restoration of lands affected by forest fires, landslides and extreme rainfall.
- Integrate ecosystem restoration into watershed development and disaster risk reduction programmes.

2. Biodiversity Conservation and Habitat Connectivity

- Strengthen landscape connectivity through restoration of wildlife corridors and ecological networks.
- Identify and conserve climate refugia for endemic and threatened species.
- Promote climate-responsive management of protected areas.
- Restore degraded alpine meadows, grasslands and high-altitude ecosystems.
- Integrate biodiversity conservation into district and landscape planning.

3. Climate-Adaptive Forest Management

- Develop climate-informed forest working plans based on projected climate scenarios.
- Promote adaptive silvicultural practices supporting species diversity and natural regeneration.
- Enhance monitoring of climate-sensitive species and ecosystem health.
- Reduce invasive alien species through integrated landscape management.
- Improve management of pest and disease outbreaks under changing climatic conditions.

4. Integrated Forest Fire Risk Management

- Expand real-time forest fire monitoring and early warning systems.
- Strengthen fuel-load management through controlled biomass removal and fire-line maintenance.
- Promote community-based fire prevention and rapid response systems.
- Enhance capacity of Van Panchayats and Forest Department staff in climate-informed fire management.

- Integrate forest fire management with ecosystem restoration programmes.

5. Ecosystem-Based Adaptation for Water and Disaster Risk Reduction

- Restore upper catchments supporting springs, rivers and drinking water sources.
- Promote nature-based slope stabilization through native vegetation.
- Restore wetlands, riparian vegetation and natural drainage systems.
- Enhance soil and moisture conservation in degraded forest landscapes.
- Integrate forest conservation with springshed and watershed management.

6. Community-Based Forest Governance and Livelihoods

- Strengthen Van Panchayats and Joint Forest Management institutions.
- Promote sustainable harvesting and value addition of Non-Timber Forest Products (NTFPs).
- Diversify livelihoods through medicinal plants, ecotourism, beekeeping and nature-based enterprises.
- Build community capacity for climate adaptation and ecosystem stewardship.
- Promote gender-responsive and inclusive forest governance.

7. Climate Information, Monitoring and Digital Innovation

- Develop long-term ecological monitoring systems for climate-sensitive ecosystems.
- Expand remote sensing and GIS-based monitoring of forest condition and landscape change.
- Deploy AI-enabled forest fire prediction and ecosystem health monitoring.
- Monitor shifts in species distribution and habitat suitability under future climate scenarios.
- Integrate forest monitoring with the Uttarakhand Digital Climate Vulnerability Atlas.

8. Institutional Strengthening and Policy Integration

- Integrate climate adaptation into Forest Working Plans and Landscape Restoration Plans.
- Strengthen coordination among Forest Department, Biodiversity Board, Disaster Management Authority, Rural Development Department and Panchayati Raj Institutions.
- Mainstream ecosystem-based adaptation into district development planning.
- Build institutional capacity for climate-informed forest management.
- Strengthen research partnerships with academic and scientific institutions for climate adaptation planning.

3.3 Tourism

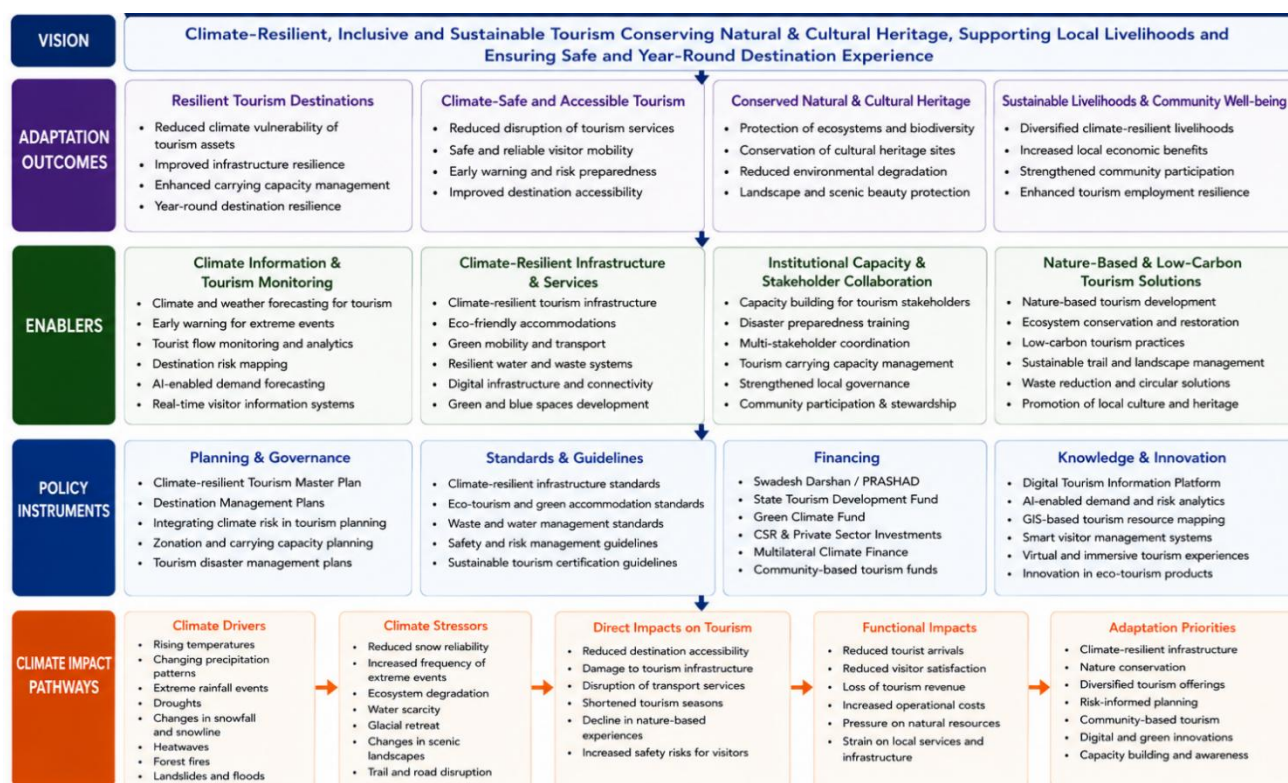
Adaptation Objective

Strengthen the resilience of Uttarakhand's tourism sector by enhancing the climate resilience of destinations, tourism infrastructure, natural and cultural assets, and tourism-dependent communities while promoting sustainable, diversified and year-round tourism under changing climatic conditions.

Adaptation Rationale

Tourism is a major contributor to Uttarakhand's economy and livelihoods, with the State's tourism industry closely linked to its mountain ecosystems, pilgrimage centres, biodiversity, adventure tourism destinations and cultural heritage. Climate change is increasingly affecting the reliability and attractiveness of these tourism assets through rising temperatures, changing snowfall patterns, extreme rainfall events, landslides, flash floods, forest fires and water scarcity.

The Climate Impact Assessment indicates that changing climatic conditions are reducing destination accessibility, increasing disruption of transport networks, affecting visitor safety, degrading natural landscapes and shortening the operational period of several tourism activities. Climate-induced degradation of glaciers, alpine ecosystems, forests and river systems may further reduce the ecological and recreational value of many destinations. Adaptation therefore requires integrated destination planning that combines climate-resilient infrastructure, ecosystem conservation, sustainable tourism management and climate-informed governance.



Strategic Adaptation Measures

1. Climate-Resilient Tourism Infrastructure

- Climate-proof tourism infrastructure including visitor centres, trekking routes, ropeways, accommodation facilities and pilgrimage infrastructure.
- Improve resilience of water supply, sanitation, solid waste management and energy systems in tourism destinations.
- Develop climate-resilient design standards for tourism facilities located in hazard-prone areas.
- Promote resilient transport infrastructure supporting year-round destination accessibility.
- Encourage green building standards and low-impact tourism infrastructure.

2. Climate-Informed Destination Planning

- Develop Climate-Resilient Destination Management Plans for major tourism destinations.
- Integrate climate risk assessment into tourism master planning and infrastructure investment.
- Establish destination carrying capacity assessments considering future climate conditions.
- Diversify tourism circuits to reduce pressure on climate-sensitive destinations.
- Promote year-round tourism by expanding nature, wellness, cultural and rural tourism.

3. Nature-Based Tourism and Ecosystem Conservation

- Restore degraded tourism landscapes through ecosystem restoration and watershed management.
- Strengthen conservation of alpine meadows, forests, wetlands and river corridors supporting tourism.
- Promote nature-based tourism linked with biodiversity conservation.
- Improve management of trekking routes and ecologically sensitive destinations.
- Integrate ecosystem-based adaptation into tourism development projects.

4. Climate Risk Management and Visitor Safety

- Establish integrated weather, landslide, flood and avalanche early warning systems for tourism destinations.
- Develop emergency preparedness and evacuation plans for major pilgrimage and tourism sites.
- Strengthen climate-informed visitor information systems.
- Improve disaster preparedness of tourism operators and local administrations.
- Establish climate risk communication protocols for tourists.

5. Sustainable Community-Based Tourism

- Promote community-based ecotourism and homestay programmes.
- Diversify tourism-based livelihoods through local products, handicrafts and cultural experiences.
- Build capacity of local tourism enterprises in climate adaptation.
- Strengthen participation of women, youth and local communities in destination management.
- Promote climate-resilient livelihood diversification in tourism-dependent regions.

6. Climate Information, Digital Tourism and Innovation

- Develop digital tourism information platforms integrating climate forecasts and hazard information.
- Deploy AI-enabled visitor flow forecasting and destination management systems.
- Expand GIS-based monitoring of tourism assets and climate-sensitive destinations.
- Develop digital decision-support systems for tourism planning.
- Strengthen monitoring of climate impacts on tourism destinations and visitor patterns.

7. Institutional Strengthening and Governance

- Integrate climate adaptation into State Tourism Policy and Destination Management Plans.
- Strengthen coordination among Tourism Department, Forest Department, Disaster Management Authority, Public Works Department, Urban Local Bodies and Panchayati Raj Institutions.
- Build institutional capacity for climate-informed tourism planning.
- Promote public-private partnerships for resilient tourism infrastructure.
- Develop monitoring frameworks for climate-resilient tourism.

3.4 Water Resources & WASH

Lead department(s): Irrigation Department, Jal Sansthan, Peyjal, Swajal Department

Adaptation Objective

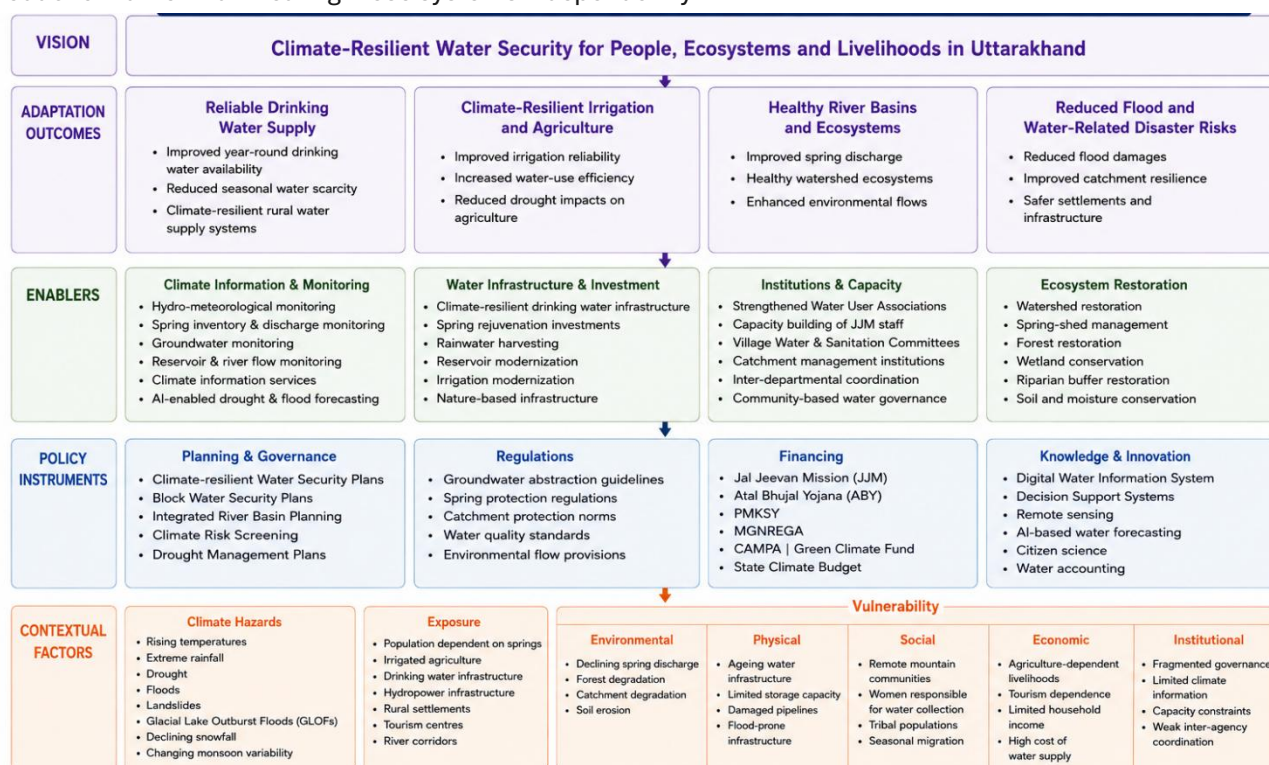
Strengthen the resilience of drinking water supply, water resources and sanitation systems to climate-induced changes in hydrological regimes, spring depletion, extreme precipitation, landslides, GLOFs and water quality

deterioration while ensuring reliable, equitable and sustainable water services for communities across Uttarakhand.

Adaptation Rationale

Spring depletion represents the dominant chronic climate risk affecting rural water security across Uttarakhand. Dependence on spring-based drinking water sources exceeds 89.1% in Bageshwar, 88.3% in Chamoli and 85.9% in Rudrapur, making these districts particularly sensitive to declining recharge, changing precipitation patterns and increasing temperatures. These long-term pressures are compounded by acute hazards including cloudbursts, floods and landslides that frequently damage gravity-fed water supply systems, while glacier-fed catchments in blocks such as Bhatwari, Joshimath, Dharchula and Munsiyari face additional risks associated with Potentially Dangerous Glacial Lakes (PDGLs) and Glacial Lake Outburst Floods (GLOFs).

Sanitation systems represent an additional but often overlooked climate vulnerability. Household sanitation infrastructure constructed under Swachh Bharat Mission (Gramin) is generally designed for service delivery rather than climate resilience. In mountainous terrain, landslides, slope instability and flash floods can rapidly damage sanitation facilities, disrupt containment systems and increase contamination risks, potentially reversing gains achieved under Open Defecation Free (ODF) programmes. Climate-resilient adaptation therefore requires integrated planning of drinking water, sanitation, watershed management and disaster risk reduction rather than treating these systems independently.



Strategic Adaptation Measures

1. Climate-Resilient Water Source Security

- Scale up springshed management using hydrogeological mapping, recharge zone protection and community-based monitoring in high spring-dependent districts.
- Protect critical recharge areas through watershed restoration, afforestation and soil-moisture conservation.

- Develop decentralized rainwater harvesting and managed aquifer recharge systems in water-stressed blocks.
- Diversify water sources to reduce dependence on single-source gravity schemes.

2. Climate-Resilient Drinking Water Infrastructure

- Climate-proof gravity-fed water supply schemes through flood-resistant intake structures, slope stabilization and resilient pipeline alignments.
- Introduce redundant water supply networks and emergency storage for critical settlements.
- Relocate vulnerable intake structures outside flood and landslide hazard zones wherever feasible.
- Integrate climate risk screening into planning and design standards under Jal Jeevan Mission.

3. Climate-Resilient Sanitation Systems

- Develop climate-resilient design standards for household and community sanitation facilities in landslide- and flood-prone areas.
- Incorporate slope stability, seismic safety and flood resilience into sanitation infrastructure guidelines.
- Promote decentralized wastewater treatment and faecal sludge management systems suitable for mountainous terrain.
- Integrate sanitation recovery planning into disaster preparedness and response mechanisms.

4. Integrated Catchment and Watershed Management

- Restore degraded catchments supporting major drinking water sources.
- Protect riparian buffers and wetlands to improve water regulation.
- Promote nature-based solutions for erosion control and groundwater recharge.
- Integrate forest restoration with springshed management and watershed development programmes.

5. Climate Information and Early Warning Systems

- Expand hydro-meteorological monitoring across priority catchments.
- Install real-time monitoring of spring discharge, groundwater levels and water quality.
- Establish flood and GLOF early warning systems for water supply infrastructure located in glacier-fed catchments.
- Integrate climate information into Water Safety Plans under Jal Jeevan Mission.

6. Institutional Strengthening and Governance

- Integrate climate risk assessment into district and block Water Security Plans.
- Strengthen coordination among Jal Jeevan Mission, Swajal, Irrigation Department, SECCCD, USDMA and Panchayati Raj Institutions.
- Build capacity of Village Water and Sanitation Committees for climate-resilient operation and maintenance.
- Develop climate-sensitive operation protocols for water supply during droughts, floods and emergency situations.

7. Emerging Adaptation Opportunities

- Digital Water Resource Management
- Deploy IoT-enabled monitoring of spring discharge, groundwater levels and drinking water quality.

- Integrate monitoring platforms with the Uttarakhand Digital Climate Vulnerability Atlas.
- Develop digital water accounting and decision-support systems for climate-informed water allocation.

8. Artificial Intelligence and Predictive Analytics

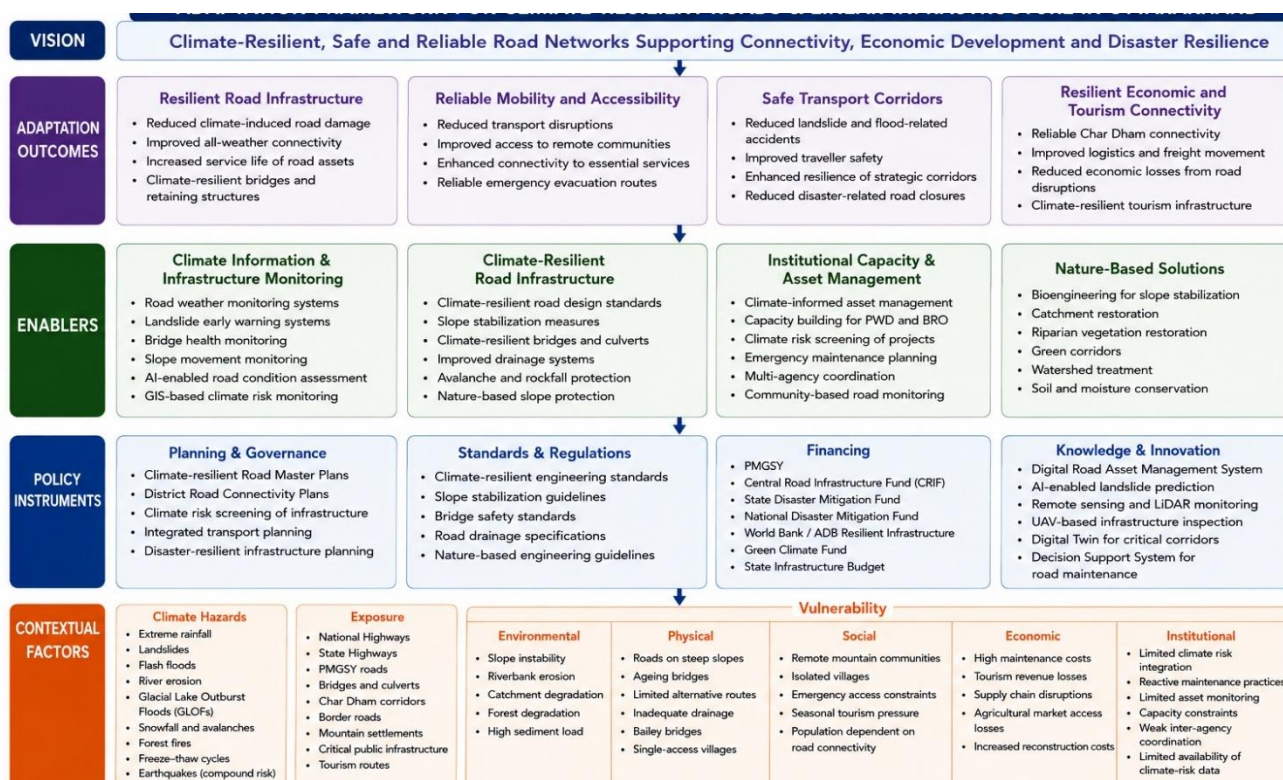
- Apply AI-based prioritisation to identify springs requiring immediate rejuvenation based on recharge potential, population served and climate risk.
- Develop machine learning models for forecasting spring discharge, drought risk and seasonal water availability.
- Use predictive analytics for optimizing maintenance of water infrastructure exposed to landslides and floods.

9. Risk-Informed Water Safety Planning

- Integrate climate hazard layers into Water Safety Plans to jointly assess water quality, quantity and climate risks.
- Combine contamination surveillance with flood, landslide and GLOF risk information for proactive management.
- Develop climate-informed emergency response protocols for rural drinking water systems.

3.5 Roads & Linear Infrastructure

Lead department(s): Public Works Department, Urban Development Department



Adaptation Objective

Strengthen the resilience of Uttarakhand's road and linear infrastructure by reducing climate-induced disruptions, improving the reliability of transport connectivity, integrating climate risk into infrastructure planning and asset management, and ensuring safe and sustainable mobility under changing climatic conditions.

Adaptation Rationale

Road networks in Uttarakhand form the backbone of economic development, tourism, pilgrimage, emergency response and border connectivity. Owing to the State's steep terrain and fragile geology, road infrastructure is highly exposed to landslides, flash floods, river erosion, cloudbursts and extreme rainfall, while climate change is expected to increase the frequency and intensity of these hazards. Disruptions to roads, bridges and associated infrastructure often isolate remote settlements, interrupt access to health and education services, delay disaster response and affect tourism and trade.

The Climate Impact Assessment indicates that increasing hydro-meteorological extremes, combined with unstable slopes, changing river morphology and sediment dynamics, are likely to increase maintenance requirements and shorten the service life of transport infrastructure. Adaptation therefore requires a transition from reactive infrastructure repair to proactive, risk-informed planning, climate-resilient engineering and integrated asset management.

Strategic Adaptation Measures

1. Climate-Resilient Road and Bridge Infrastructure

- Integrate projected climate scenarios into the planning and design of roads, bridges, tunnels and retaining structures.
- Upgrade drainage systems, culverts and cross-drainage structures to accommodate projected increases in extreme rainfall.
- Strengthen slope stabilisation using engineering and bioengineering approaches in landslide-prone corridors.
- Climate-proof bridges and river crossings against increased flood intensity, erosion and sediment transport.
- Adopt resilient pavement materials and construction practices suitable for freeze–thaw conditions and changing climatic regimes.

2. Climate Risk-Informed Infrastructure Planning

- Conduct climate risk screening for all new road and bridge projects.
- Incorporate hazard zonation and future climate projections into route selection and infrastructure planning.
- Prioritise investments in critical transport corridors connecting remote communities, pilgrimage routes and strategic border areas.
- Develop alternative access routes and redundancy for highly vulnerable transport corridors.
- Mainstream climate adaptation into State and district transport master plans.

3. Intelligent Asset Management and Monitoring

- Establish digital road asset management systems incorporating climate risk information.
- Deploy remote sensing, LiDAR and UAV-based monitoring of slopes, bridges and road corridors.
- Introduce structural health monitoring for critical bridges and retaining structures.
- Develop predictive maintenance systems based on climate forecasts and asset condition monitoring.
- Maintain geospatial inventories of climate-sensitive transport assets.

4. Climate Risk Management and Emergency Preparedness

- Strengthen multi-hazard early warning systems for landslides, floods and extreme rainfall affecting transport corridors.
- Prepare emergency response and rapid restoration plans for critical road networks.
- Improve availability of emergency equipment and strategic maintenance depots.
- Develop standard operating procedures for road closures, traffic diversion and emergency access.
- Strengthen coordination between transport agencies and disaster management authorities during extreme events.

5. Nature-Based Solutions for Slope and Catchment Stability

- Restore degraded catchments contributing to road instability.
- Promote bioengineering techniques for slope stabilisation using native vegetation.
- Restore riparian vegetation to reduce riverbank erosion affecting transport corridors.
- Implement watershed management and soil conservation measures upstream of vulnerable road sections.
- Integrate ecosystem-based adaptation into linear infrastructure planning.

6. Climate Information, Digital Technologies and Innovation

- Develop AI-enabled landslide susceptibility forecasting and road closure prediction systems.
- Expand road weather monitoring networks along major transport corridors.
- Integrate transport infrastructure into the Uttarakhand Digital Climate Vulnerability Atlas.
- Establish GIS-based decision-support systems for maintenance prioritisation.
- Develop digital twin platforms for critical transport corridors to support scenario-based planning and asset management.

7. Institutional Strengthening and Capacity Building

- Mainstream climate resilience into Public Works Department (PWD), Border Roads Organisation (BRO) and National Highways planning processes.
- Strengthen coordination among PWD, BRO, USDMA, SECCCD, district administrations and local governments.
- Develop technical guidelines for climate-resilient road engineering in mountain environments.
- Build institutional capacity in climate risk assessment, resilient infrastructure design and adaptive asset management.
- Promote research partnerships with academic and technical institutions to improve climate-resilient infrastructure planning.

3.6 Energy

Lead department(s): Uttarakhand Power Corporation Ltd, Power Transmission Corporation of Uttarakhand

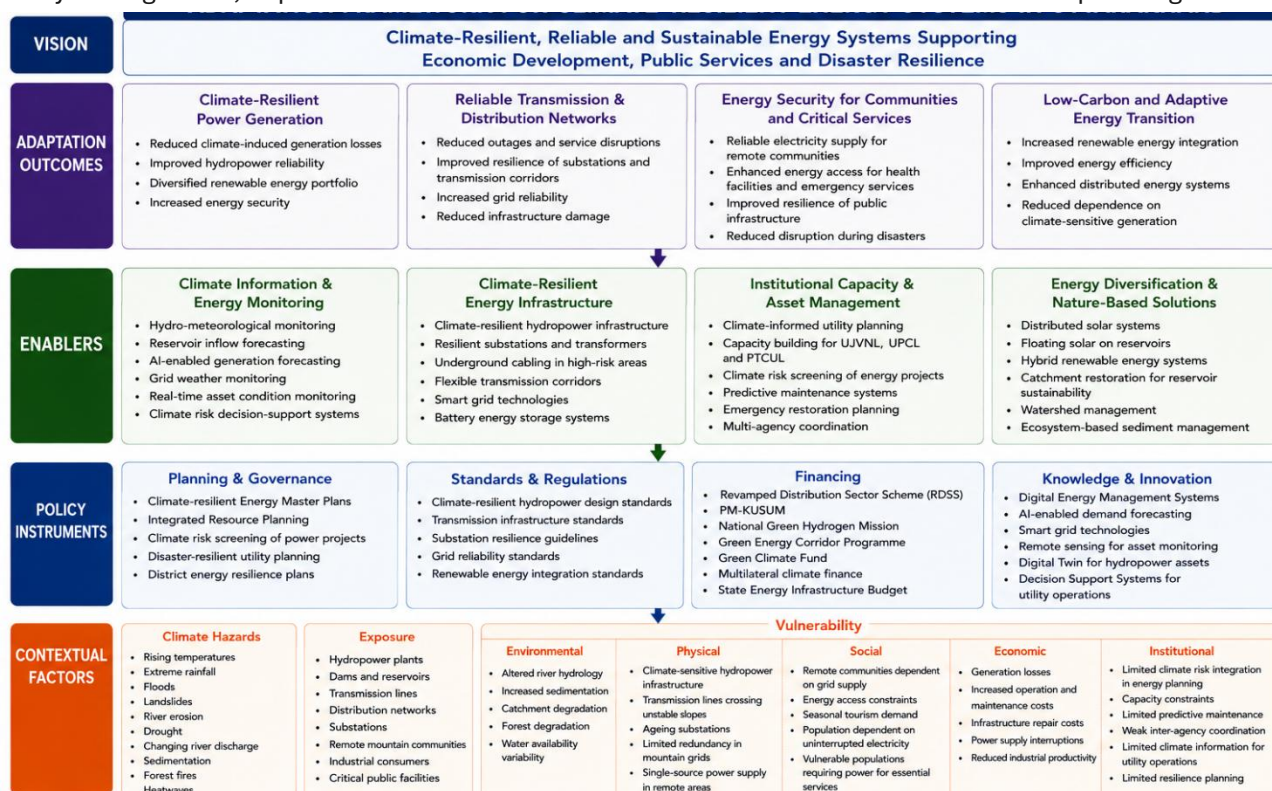
Adaptation Objective

Strengthen the resilience of Uttarakhand's energy sector by enhancing the reliability of hydropower generation, climate-proofing transmission and distribution infrastructure, diversifying renewable energy sources, and integrating climate risk into energy planning and utility management to ensure secure and sustainable electricity supply under changing climatic conditions.

Adaptation Rationale

The energy sector in Uttarakhand is highly dependent on hydropower, making electricity generation intrinsically linked to changes in river hydrology, glacier-fed flows and catchment conditions. Climate change is expected to alter seasonal runoff patterns, increase hydrological variability and accelerate sediment transport, affecting reservoir operations, turbine efficiency and long-term generation reliability. Extreme rainfall, flash floods, landslides and river erosion are increasingly damaging hydropower infrastructure, substations and transmission corridors, while prolonged dry periods may reduce generation during lean-flow seasons. In mountainous terrain, disruption of transmission networks can isolate remote communities and interrupt electricity supply to critical services, including healthcare, water supply and emergency response.

The Climate Impact Assessment indicates that strengthening energy resilience requires an integrated approach encompassing climate-resilient infrastructure, diversified renewable energy systems, adaptive utility management, improved climate information services and risk-informed investment planning.



Strategic Adaptation Measures

1. Climate-Resilient Hydropower Generation

- Integrate climate projections into hydropower planning, design and operational protocols.
- Modernise hydropower facilities to improve operational flexibility under changing flow regimes.
- Strengthen sediment management through desilting systems, catchment treatment and reservoir operation optimisation.
- Enhance climate-resilient dam safety monitoring and emergency preparedness.
- Develop adaptive reservoir operation guidelines considering projected hydrological variability.

2. Climate-Resilient Transmission and Distribution Infrastructure

- Climate-proof substations, transformers and transmission corridors exposed to floods, landslides and river erosion.
- Strengthen slope stabilisation and protective engineering around transmission infrastructure.

- Underground distribution lines in high-risk urban and critical service areas, where technically feasible.
- Increase redundancy within mountain electricity networks to minimise service interruptions.
- Develop resilient power supply systems for remote settlements and critical infrastructure.

3. Renewable Energy Diversification and Distributed Energy Systems

- Expand decentralised solar photovoltaic systems for remote and disaster-prone communities.
- Promote hybrid renewable energy systems combining solar, small hydropower and battery storage.
- Support rooftop solar installations for public buildings, health facilities and educational institutions.
- Encourage floating solar installations on suitable reservoirs where environmentally appropriate.
- Promote community-scale renewable energy systems to improve local energy security.

4. Climate Risk Management and Utility Resilience

- Develop utility-specific climate risk management plans.
- Establish emergency restoration protocols for climate-induced infrastructure damage.
- Improve disaster preparedness for hydropower stations and transmission utilities.
- Strengthen continuity planning for critical facilities including hospitals, water supply systems and emergency response centres.
- Develop contingency planning for prolonged low-flow periods and extreme weather events.

5. Climate Information and Smart Energy Systems

- Establish hydro-meteorological monitoring systems supporting reservoir operations.
- Develop AI-enabled inflow forecasting and hydropower generation forecasting.
- Deploy digital asset management systems for predictive maintenance.
- Expand real-time monitoring of substations, transmission lines and hydropower infrastructure.
- Integrate climate information into utility decision-support systems.

6. Catchment Management and Ecosystem-Based Adaptation

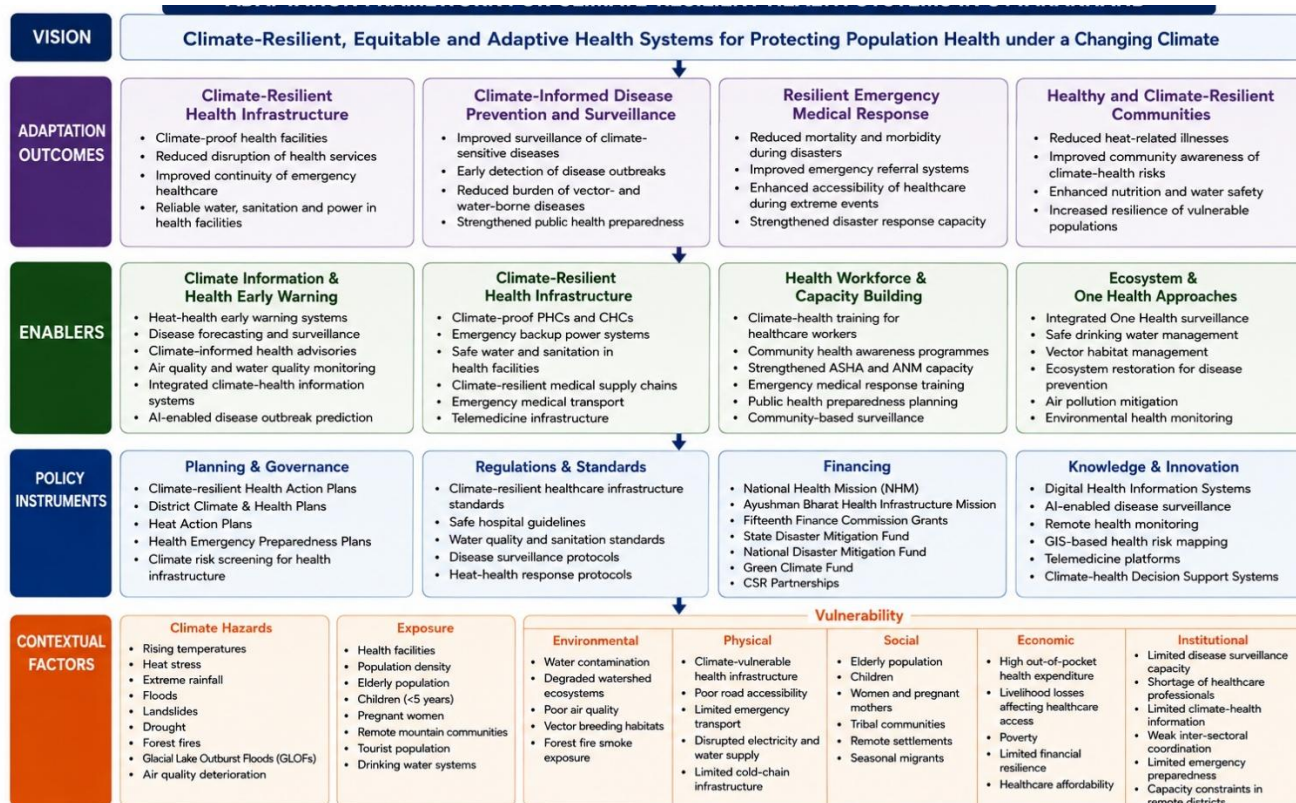
- Objective: Strengthen ecosystem services supporting long-term hydropower sustainability.
- Priority Actions
 - Restore degraded catchments upstream of major hydropower reservoirs.
 - Promote watershed restoration to reduce erosion and sediment loads.
 - Stabilise riverbanks and slopes using nature-based solutions.
 - Integrate forest restoration with catchment conservation programmes.
 - Promote ecosystem-based approaches for improving long-term water regulation.

7. Institutional Strengthening and Climate-Responsive Energy Governance

- Mainstream climate risk assessment into State Energy Plans and utility investment planning.
- Strengthen coordination among UJVNL, UPCL, PTCUL, USDMA, SECCCD and other relevant agencies.
- Build institutional capacity for climate-resilient utility management.
- Integrate climate screening into appraisal of future hydropower and transmission projects.
- Develop climate-responsive operation and maintenance protocols for energy infrastructure.

3.7 Health

Lead department(s): Department of Health & Family Welfare, State Health Mission



Adaptation Objective

Strengthen the resilience of Uttarakhand's health system by enhancing climate-informed disease surveillance, climate-resilient health infrastructure, emergency preparedness, and equitable access to essential health services while reducing the health impacts of climate variability and extreme weather events.

Adaptation Rationale

Climate change is increasingly affecting public health in Uttarakhand through rising temperatures, changing precipitation patterns, extreme rainfall, floods, landslides, forest fires and deteriorating water quality. These hazards are expected to increase the incidence of heat-related illnesses, water- and vector-borne diseases, respiratory illnesses associated with forest fire smoke, injuries during disasters and disruptions to healthcare services. Remote mountain communities remain particularly vulnerable due to difficult terrain, limited healthcare accessibility and disruptions to transport and communication during extreme weather events.

The Climate Impact Assessment indicates that increasing climate variability is likely to place additional pressure on health infrastructure, disease surveillance systems and emergency medical services. Strengthening resilience therefore requires an integrated approach that combines climate-informed

public health planning, resilient healthcare infrastructure, digital health technologies, early warning systems and cross-sectoral coordination.

Strategic Adaptation Measures

1. Climate-Resilient Health Infrastructure

- Climate-proof Primary Health Centres (PHCs), Community Health Centres (CHCs), district hospitals and critical healthcare infrastructure against floods, landslides and extreme weather.
- Ensure uninterrupted electricity, safe drinking water and sanitation facilities in health institutions through resilient utility systems and backup power.
- Improve structural safety of health facilities located in hazard-prone areas.
- Develop climate-resilient medical supply chains and emergency logistics systems.
- Promote green and energy-efficient healthcare infrastructure.

2. Climate-Informed Disease Surveillance and Early Warning

- Integrate climate information into surveillance of vector-borne, water-borne and respiratory diseases.
- Strengthen early warning systems for heatwaves, disease outbreaks and air quality-related health risks.
- Expand environmental health surveillance for drinking water quality, sanitation and food safety.
- Develop predictive disease modelling using meteorological and epidemiological data.
- Strengthen laboratory capacity for rapid diagnosis and outbreak detection.

3. Heat and Disaster Health Preparedness

- Develop district-level Heat Action Plans integrating health sector preparedness.
- Strengthen emergency medical response for floods, landslides, forest fires and other climate-induced disasters.
- Improve emergency referral networks for remote mountain communities.
- Establish contingency plans for mass casualty incidents during disasters and pilgrimage seasons.
- Conduct regular preparedness exercises and emergency response training for healthcare personnel.

4. Community Health Resilience and Risk Communication

- Strengthen climate and health awareness programmes at community level.
- Promote safe drinking water, hygiene and sanitation practices in climate-vulnerable communities.
- Enhance nutrition programmes for vulnerable populations during climate-related emergencies.
- Strengthen community-based disease surveillance through ASHAs, ANMs and Village Health, Sanitation and Nutrition Committees (VHSNCs).
- Develop targeted risk communication strategies for heat stress, floods, vector-borne diseases and poor air quality.

5. Climate Information, Digital Health and Innovation

- Establish integrated Climate and Health Information Systems linking meteorological, environmental and health data.
- Deploy AI-enabled disease forecasting and outbreak prediction models.

- Expand telemedicine services to improve healthcare access in remote and disaster-prone areas.
- Develop GIS-based health vulnerability mapping and decision-support systems.
- Strengthen digital health records to support climate-sensitive health surveillance and response.

6. One Health and Environmental Health Management

- Strengthen One Health surveillance for zoonotic and climate-sensitive diseases.
- Improve coordination between the Health, Animal Husbandry, Forest and Environment Departments.
- Monitor environmental determinants of health, including water quality, air quality and vector habitats.
- Promote ecosystem-based approaches to reduce vector breeding and improve environmental health.
- Integrate environmental monitoring into public health planning.

7. Institutional Strengthening and Climate-Responsive Health Governance

- Integrate climate risk assessment into State and District Health Action Plans.
- Strengthen institutional coordination among the Health Department, National Health Mission (NHM), USDMA, SECCCD, IMD and local governments.
- Build capacity of healthcare professionals on climate-sensitive health risks and emergency response.
- Develop climate-responsive health financing and investment strategies.
- Strengthen monitoring and evaluation of climate adaptation interventions within the health sector.

4. Adaptation Recommendation Measures: Block-wise Table (95 Blocks)

The table on the following pages operationalises the sector recommendations of at the level of all 95 blocks across Uttarakhand's 13 districts:

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
Uttarkashi	Bhatwari	Landslide + Spring/Hydropower	Very High	High	Very High	Moderate	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by spring & source depletion; hydropower / transmission exposure (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-proofing of intake/desilting structures; GLOF-linked automatic shutdown protocol	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; UPCL / UJVNL
Uttarkashi	Dunda	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Uttarkashi	Chinyalisaur	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
Uttarkashi	Naugaon	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Uttarkashi	Purola	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Uttarkashi	Mori	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Chamoli	Dasholi	Landslide + Chilling/Spring/Hydropower/Vector	Very High	Very High	Very High	Very High	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress; spring & source depletion; hydropower / transmission exposure; vector- & waterborne-disease burden (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-resilient	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; Dept. of Horticulture

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
										orchard package: low-chill cultivar substitution, high-density replanting, protective netting		
Chamoli	Joshimath	Landslide + Chilling/Spring/Hydropower/Vector	Very High	Very High	Very High	Very High	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress; spring & source depletion; hydropower / transmission exposure; vector- & waterborne-disease burden (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; Dept. of Horticulture
Chamoli	Ghat	Landslide + Vector	High	High	High	Very High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by vector- & waterborne-disease burden (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Climate-informed vector surveillance and last-mile cold-chain/health outreach	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO; Health & Family Welfare
Chamoli	Karnaprayag	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure	Slope stabilisation (bioengineering + retaining structures) on priority failure-	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									corridors (composite terrain exposure: High).	prone stretches; Landslide early-warning sensors on critical corridors		
Chamoli	Dewal	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Chamoli	Tharali	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Chamoli	Narayanbagar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Chamoli	Pokhri	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Chamoli	Gairsain	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements,	Slope stabilisation (bioengineering + retaining	Catchment afforestation and terrace stabilisation	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									access roads and infrastructure corridors (composite terrain exposure: Moderate).	structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	(carbon sink + slope co-benefit)	
Rudraprayag	Ukhimath	Landslide + Spring/Hydropower/Vector	Very High	High	Very High	Very High	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by spring & source depletion; hydropower / transmission exposure; vector- & waterborne-disease burden (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-proofing of intake/desilting structures; GLOF-linked automatic shutdown protocol	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; UPCL / UJVNL
Rudraprayag	Agastyamuni	Landslide	Very High	High	High	Moderate	Very High	High	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Priority inclusion in the Digital Vulnerability Atlas early-action tier	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Rudraprayag	Jakholi	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure	Slope stabilisation (bioengineering + retaining structures) on priority failure-	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									corridors (composite terrain exposure: High).	prone stretches; Landslide early-warning sensors on critical corridors		
Tehri Garhwal	Chamba	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Jaunpur	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Kirtinagar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Narendranagar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Pratapnagar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements,	Slope stabilisation (bioengineering + retaining	Catchment afforestation and terrace stabilisation	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									access roads and infrastructure corridors (composite terrain exposure: High).	structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	(carbon sink + slope co-benefit)	
Tehri Garhwal	Bhilangana	Landslide	Very High	High	High	Moderate	Very High	High	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Priority inclusion in the Digital Vulnerability Atlas early-action tier	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Jakhnidhar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Thauldhar	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Tehri Garhwal	Devprayag	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
										on critical corridors		
Dehradun	Chakrata	Extreme Rainfall	High	High	High	High	High	Moderate	Cloudburst-driven slope failure, flash discharge and infrastructure disruption (composite terrain exposure: High).	Cloudburst-resilient drainage and culvert redesign; Automatic rain-gauge densification and threshold-based alerts	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	Revenue Dept. / USDMA
Dehradun	Kalsi	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Dehradun	Vikasnagar	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Dehradun	Sahaspur	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Dehradun	Raipur	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
Dehradun	Doiwala	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Pauri Garhwal	Kot	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Kaljikhal	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Pauri	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Pabau	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
Pauri Garhwal	Thalisain	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Bironkhal	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Dwarikhal	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Ekeshwar	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Jaharikhal	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
										on critical corridors		
Pauri Garhwal	Nainidanda	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Pokhra	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Rikhnikhal	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Yamkeshwar	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pauri Garhwal	Dugadda	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches;	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									(composite terrain exposure: High).	Landslide early-warning sensors on critical corridors		
Pauri Garhwal	Khirsu	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Nainital	Betalghat	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Nainital	Bhimtal	Landslide + Chilling/Hydropower	High	Very High	High	High	High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress; hydropower / transmission exposure (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting; Climate-proofing of intake/desilting structures; GLOF-linked automatic shutdown protocol	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Small-hydro efficiency retrofits reduce sedimentation-linked generation loss	PWD / BRO; Dept. of Horticulture; UPCL / UJVNL
Nainital	Dhari	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and	Slope stabilisation (bioengineering + retaining structures) on	Catchment afforestation and terrace stabilisation	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									infrastructure corridors (composite terrain exposure: High).	priority failure-prone stretches; Landslide early-warning sensors on critical corridors	(carbon sink + slope co-benefit)	
Nainital	Haldwani	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Nainital	Kotabag	Flood	High	High	High	High	High	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: High).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Nainital	Okhalkanda	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Nainital	Ramgarh	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Nainital	Ramnagar	Flood	High	High	High	High	High	Moderate	Riverine/flash-flood exposure of valley-floor settlements,	Flood-proofing of intakes, embankments and low-lying	Wetland/floodplain restoration for natural attenuation	Irrigation Dept. / USDMA

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									schemes and croplands (composite terrain exposure: High).	settlements; River-stage gauges linked to community alert protocol	and biodiversity co-benefit	
Almora	Hawalbagh	Landslide + Chilling	Moderate	Very High	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Agroforestry buffer strips support pollinator habitat and orchard microclimate	PWD / BRO; Dept. of Horticulture
Almora	Bhaisiyachhana	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Bhikiyasain	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Chaukhutia	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
										on critical corridors		
Almora	Dhauladevi	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Dwarahat	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Lamgara	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Sult	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Syaldey	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches;	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									(composite terrain exposure: High).	Landslide early-warning sensors on critical corridors		
Almora	Takula	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Almora	Tarikheth	Landslide	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: Moderate).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Bageshwar	Bageshwar	Landslide + Vector	High	High	High	Very High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by vector- & waterborne-disease burden (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Climate-informed vector surveillance and last-mile cold-chain/health outreach	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO; Health & Family Welfare
Bageshwar	Garur	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
Bageshwar	Kapkot	Landslide + Chilling/Spring/Hydropower/Vector	Very High	Very High	Very High	Very High	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress; spring & source depletion; hydropower / transmission exposure; vector- & waterborne-disease burden (composite terrain exposure: Very High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; Dept. of Horticulture
Champawat	Champawat	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Champawat	Lohaghat	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Champawat	Pati	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure	Slope stabilisation (bioengineering + retaining structures) on priority failure-	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									corridors (composite terrain exposure: High).	prone stretches; Landslide early-warning sensors on critical corridors		
Champawat	Barakot	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Berinag	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Didihat	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Dharchula	Landslide + Chilling/Spring/Hydropower/Vector	Very High	Very High	Very High	Very High	Very High	Very High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit / orchard-livelihood stress; spring & source depletion; hydropower / transmission exposure; vector-	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors; Springshed management under the NITI Aayog 8-step	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces sediment load and supports downstream irrigation	PWD / BRO; Jal Sansthan / Swajal; Dept. of Horticulture

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									& waterborne-disease burden (composite terrain exposure: Very High).	protocol (recharge-zone protection, spring inventory, community stewardship); Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting		
Pithoragarh	Gangolihat	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Kanalichhina	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Munakot	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Pithoragarh	Munsyari	Landslide + Chilling/Spring/Vector	Very High	Very High	Very High	Very High	Very High	High	Slope instability threatening settlements, access roads and infrastructure corridors; compounded by chilling deficit /	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit); Recharge-forestry and check-dam network reduces	PWD / BRO; Jal Sansthan / Swajal; Dept. of Horticulture

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									orchard-livelihood stress; spring & source depletion; vector- & waterborne-disease burden (composite terrain exposure: Very High).	on critical corridors; Springshed management under the NITI Aayog 8-step protocol (recharge-zone protection, spring inventory, community stewardship); Climate-resilient orchard package: low-chill cultivar substitution, high-density replanting, protective netting	sediment load and supports downstream irrigation	
Pithoragarh	Vin (Berinag)	Landslide	High	High	High	High	High	Moderate	Slope instability threatening settlements, access roads and infrastructure corridors (composite terrain exposure: High).	Slope stabilisation (bioengineering + retaining structures) on priority failure-prone stretches; Landslide early-warning sensors on critical corridors	Catchment afforestation and terrace stabilisation (carbon sink + slope co-benefit)	PWD / BRO
Haridwar	Bahadrabad	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Haridwar	Bhagwanpur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Haridwar	Khanpur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements,	Flood-proofing of intakes, embankments and low-lying	Wetland/floodplain restoration for natural attenuation	Irrigation Dept. / USDMA

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									schemes and croplands (composite terrain exposure: Low).	settlements; River-stage gauges linked to community alert protocol	and biodiversity co-benefit	
Haridwar	Laksar	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Haridwar	Narsan	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Haridwar	Roorkee	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Bazpur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Gadarpur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA

District	Block	Priority Hazard	Risk Score	AVI	WVI	HVI	RVI	EVI	Key Vulnerability	Adaptation Measures	Mitigation Co-benefit	Lead Dept.
									(composite terrain exposure: Low).	community alert protocol		
Udham Singh Nagar	Jaspur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Kashipur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Khatima	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Rudrapur	Flood	Low	Low	Low	Low	Low	Low	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Low).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA
Udham Singh Nagar	Sitarganj	Flood	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate	Riverine/flash-flood exposure of valley-floor settlements, schemes and croplands (composite terrain exposure: Moderate).	Flood-proofing of intakes, embankments and low-lying settlements; River-stage gauges linked to community alert protocol	Wetland/floodplain restoration for natural attenuation and biodiversity co-benefit	Irrigation Dept. / USDMA

Table 4.1 Block-wise adaptation and mitigation recommendation matrix, all 95 blocks, 13 districts.

5. Mainstreaming Climate Risk into Development Planning

5.1 Purpose

Mainstreaming ensures that the hazard, exposure and vulnerability findings of this assessment do not remain a stand-alone technical output but are embedded in the recurring planning, budgeting and monitoring cycles of the State, its 13 districts and its 95 blocks. The objective is convergence with, not duplication of, existing planning instruments into the State already runs SAPCC/UAPCC, SAPCCHH, District Disaster Management Plans (DDMPs), District Irrigation Plans and Gram Panchayat Development Plans (GPDPs); this report's outputs are designed to plug directly into each of these cycles.

5.2 Mainstreaming into State Plans

- Update the 2014 UAPCC/SAPCC using this report's block-level hazard and vulnerability evidence base, aligning sectoral targets with national net-zero-by-2070 commitments and the eight NAPCC missions.
- Embed the Digital Vulnerability Atlas as a standing layer within SECCCD's State Climate Change Centre so hazard, exposure and vulnerability data refresh on a defined cycle rather than as a one-time output.
- Require climate-risk screening (using this report's block matrix) as a standard input to State-sector annual plans and capital-works sanctioning, beginning with PWD, Irrigation, UPCL and Health & Family Welfare.

5.3 Mainstreaming into District Plans

- Integrate block-wise priority-hazard and vulnerability data into District Disaster Management Plans (DDMPs) under the Disaster Management Act, 2005, refreshed annually ahead of the monsoon season.
- Align District Irrigation Plans and District Health Action Plans with the sector-specific measures in Section 3, prioritised by the Very-High composite blocks identified in Section 4.
- Establish a District Climate Cell (or formally task the existing District Disaster Management Authority) with responsibility for tracking block-level adaptation progress against the monitoring indicators in Section 9.

5.4 Mainstreaming into Block Plans

- Incorporate block-specific adaptation measures from Section 4 into Gram Panchayat Development Plans (GPDPs) and block-level MGNREGA labour-budget planning, particularly for springshed treatment, afforestation, slope stabilisation and water-harvesting works that are MGNREGA-eligible.
- Use Van Panchayats and Jal Sansthan/Swajal block units as the primary community-level implementation and monitoring nodes for forest, water and WASH measures respectively.

5.5 Convergence with Existing Government Schemes

Scheme / Programme	Convergence opportunity
Jal Jeevan Mission (JJM) / Swajal	Climate-proof scheme design (flood/landslide-resilient intakes); springshed management as a JJM source-sustainability input
MGNREGA	Labour-budget financing for afforestation, terracing, check-dams and slope-stabilisation works identified in the block matrix
PMFBY (crop insurance) & PM-KISAN	Fast-tracked claims for extreme-weather and human-wildlife-conflict crop loss in priority agriculture blocks
AMRUT 2.0 / Building Heat Resilient Cities	Urban heat and drainage measures in Dehradun, Haridwar, Udham Singh Nagar plains blocks

Scheme / Programme	Convergence opportunity
CAMPA (Compensatory Afforestation Fund)	Financing for catchment afforestation, springshed forestry and forest-fire-line management
National Health Mission (NHM) / SAPCCHH	Climate-sensitive disease surveillance and health-facility resilience retrofits
Char Dham corridor / PMGSY road works	Bioengineering and drainage retrofits sequenced into scheduled road-widening and maintenance cycles

Table 5.1 Convergence pathways with existing central and state schemes.

6. Financing Opportunities

No single financing instrument can carry the full programme implied by Sections 3 and 4. The recommended approach blends State and Central budgetary allocations for near-term, high-priority works with climate-dedicated and international finance for larger structural investments (springshed programmes, GLOF early-warning networks, health-system resilience), and private/CSR and insurance instruments for risk-transfer and livelihood-diversification measures.

6.1 Public Finance

- State Plan and Central Sector Scheme allocations routed through the sectoral departments identified as lead in Sections 3–4 (PWD, Irrigation, UPCL, Health & Family Welfare, Forest Department, Agriculture/Horticulture, Tourism Development Board).
- Fifteenth/Sixteenth Finance Commission climate-resilience grants to local bodies, tied to the district- and block-level mainstreaming pathway in Section 5.
- District Mineral Foundation (DMF) and CAMPA funds for afforestation, springshed and slope-stabilisation works in mining- and forest-adjacent blocks.

6.2 Climate-Dedicated and International Finance

- National Adaptation Fund for Climate Change (NAFCC) and the Green Climate Fund (GCF), appropriate for GLOF early-warning infrastructure, springshed-management scale-up and health-system climate-resilience investments requiring multi-year, multi-crore financing.
- Adaptation Fund and bilateral technical cooperation (GIZ, KfW, SDC/ICIMOD partnerships already active on springshed management in Uttarakhand) for capacity-building and pilot-to-scale programmes.
- World Bank / ADB disaster-resilience lending lines, appropriate for large infrastructure retrofits (transmission, substations, strategic road corridors) in Very-High composite blocks.

6.3 Private Sector, Insurance and Community Finance

- Corporate Social Responsibility (CSR) financing for community-level measures (fire-watch equipment, solar cold-chain units, homestay climate-proofing) where State budget cycles are slower than the need.
- Parametric/index-based insurance for agriculture (linked to IMD gridded thresholds) and tourism-sector business-interruption cover tied to verified road-closure or weather triggers.
- Payments for Ecosystem Services (PES) and carbon-finance instruments linked to the State's Gross Environment Product Index, to sustain community-based fire management and afforestation beyond initial public financing.

7. Institutional Framework

SECCCD is the nodal department for climate risk in Uttarakhand and is best placed to coordinate implementation of this report's recommendations, but delivery is necessarily distributed across the sectoral departments identified as lead in Section 3. The framework below sets out coordination, not command-and-control, relationships.

7.1 State-Level Coordination

- SECCCD (nodal): owns the Digital Vulnerability Atlas, coordinates cross-sectoral prioritisation, and reports progress to the State Climate Change Centre.
- Uttarakhand State Disaster Management Authority (USDMA): co-owns hazard early-warning systems (GLOF, landslide, extreme rainfall) and cross-cutting disaster-response protocols.
- India Meteorological Department (IMD, Dehradun): supplies the climate data backbone (gridded rainfall/temperature, forecasts) underpinning all sectoral early-warning systems.
- Sectoral departments (PWD, Irrigation/Jal Sansthan/Swajal, UPCL/UJVNL, Forest Department, Agriculture/Horticulture, Health & Family Welfare, Tourism Development Board): own sector-specific implementation per Section 3.

7.2 District and Block Level

- District Disaster Management Authorities (DDMAs): integrate block-level vulnerability data into DDMPs and coordinate district-level emergency response.
- District Climate Cells (proposed, or an expanded DDMA mandate): track block-level adaptation-measure implementation against the Section 9 monitoring framework.
- Van Panchayats, Water User Associations and Gram Panchayats: primary community-level implementation partners for forest, water/WASH and agriculture measures.

7.3 Technical and Knowledge Partners

8. Implementation Roadmap

Implementation is phased over three horizons, sequenced to front-load measures in the Very-High composite blocks identified in Section 4 while building the institutional and data foundations (Digital Vulnerability Atlas, monitoring framework) needed to sustain the programme beyond the initial phase.

9. Monitoring Indicators

Monitoring tracks both process (are measures being implemented as planned) and outcome (is vulnerability reducing) at block, district and state level, feeding back into periodic revision of the Digital Vulnerability Atlas and this report's recommendations.

9.1 Process Indicators

9.2 Outcome Indicators

- Change in composite risk band (Very High → High → Moderate) for Section 4 priority blocks, re-scored against the full indicator workbook once available.
- Reduction in ODF-reversion incidents following extreme-weather events in the highest ODF-gap districts (Haridwar, Almora, Rudraprayag, Pauri Garhwal, Udham Singh Nagar).
- Trend in orchard yield and chilling-hour-deficit-linked crop loss in Almora, Chamoli, Nainital, Bageshwar and Pithoragarh.
- Vector-borne and waterborne disease incidence trend in Chamoli, Rudraprayag, Bageshwar and the newly surveyed border blocks (Dharchula, Munsyari).

9.3 Data Architecture

Monitoring data should route through the same Digital Vulnerability Atlas platform proposed for hazard and vulnerability visualisation, with a defined update cadence (real-time for early-warning telemetry; seasonal for agriculture/health surveillance; annual for infrastructure-retrofit progress and composite risk re-scoring), so that the M&E function and the planning function draw on one authoritative dataset rather than parallel reporting systems.

10. Conclusion
